KIRLOSKAR CHILLERS PRIVATE LIMITED
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ENGINEERING BULLETIN

Subject :	Brine chillers operating range & other details.				
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This bulletin announces the minimum brine temperatures which can be offered for R-134a & 407C screw brine chillers. Along with it, requirement of oil cooler and constraints for dual mode operation are clearly specified. The information in this document supersedes wherever conflicting with earlier released rating charts, datasheet etc.

1) **Operating Range:**

Refrigerant	Minimum brine temp. (Evap. Leaving)	Min^{m} Compressor Unloading (For I Ckt.)	
	deg. C	%	
R134a	-10	25%	
R407C	-20	Up to -15	25%
		Below -15	50%

Notes:

A. Above table applies to all water cooled screw chiller series.
B. Minimum unloading % mentioned above are for single compressor that means for 407C chiller with double compressor, capacity control will be up to 25%.
C. Earlier released rating chart for 134a brine, ESP/13-14/102 & ESP/14-15/03 will be obsolete hereafter.

2) **Oil Cooler Requirement:**

A. As earlier up to -10 deg. C, no oil cooler is required for both R134a & R407C.
B. Below -10 deg C for R407C, revised rating chart ESP/13-14/103 & ESP/14-15/04 (accompanying with bulletin) should be referred to know oil cooler requirement. Conditions for which oil cooler is required are marked in grey color.
C. In case of Economizer, Oil cooler is must for all conditions below -10 deg. C.
D. Oil cooler cost should be considered for above point B&C.

3) **Dual Mode Brine Chillers:**

A. Dual mode brine chiller means chiller with option of 2 target temperatures, either of which can be selected on controller by end user. These can be in any combinations like +/+, +/- or -/-, provided constraints mentioned below are met.
B. There is limitation on maximum difference between two target temperatures depending on minimum temp. Required as below.

Minimum target temp. out of two in ^{o}C	Max. allowable diff. between two modes in ^{o}C
up to -5	12
up to -10	10
below -10	5

For example, if one of the temperature requirement is of -4 deg C, the other temperature can be maximum 8 deg. C (Since up to -5, diff could be max 12). Following above constraints performance of dual mode chiller should be taken using selection software and/or rating chart.

C. Up to -5 deg. C, design evaporator flows can be different for two modes calculated based on corresponding performance or higher flow can be used for both the modes then accordingly delta T at lower temp. will modify.
D. Below -5 deg. C, compulsorily higher flow (corresponding to higher temp.) should be considered for both the modes. Accordingly delta T for lower temp. should be calculated.
E. Chiller operating at 3 modes shall not be offered.